

Segmentation of fish chromosomes in microscopy images: A new Dataset

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Abstract—

Index Terms—Fish karyotype chromosomes, Computational vision, digital image.

I. INTRODUCTION

The karyotype assembly is an important task in the cytogenetics, it is useful in various practical and research activities, it aids in the diagnosis of genetic diseases, some cancer types and cytogenetic research [1]. The chromosomes are composed of supercoiled and associated DNA. Human chromosomes can suffer related anomalies to an atomic number of chromosomes or structural abnormality in one or more chromosomes [2]. The human cells contains 46 chromosomes including 22 pairs of chromosome and two sex chromosomes (XY: male and XX: female). Fish have variations in the number of chromosomes and can not be previously defined as of humans [3]. The process of chromosomal karyotyping is made by pairing the chromosomes according to the similarity between them. The chromosomes are classified in one of the 4 classes: metacentric, sub mutation, subteloc input and acrocentric according to the location of the centromere [4] [5].

In the process of segmentation, the taken images are first converted in binary format. The binary image helps find details about the shape of any object [6]. Chromosomes are cellular structures that contain genetics in formation. When chromosomes are imaged by microscopy, information on the health of an individual. Since the 1980s, chromosome detection systems and automatic classification has aroused great interest of research. A large number of studies have been carried out, to develop detection and classification systems of chromosomes. The manual assembly of the karyotype is a repetitive task, exhausting, time consuming and subject to errors, can be performed by means of visual analysis and requires the attention of an experienced and specialized professional. In some cases the analysis of many metastasis and needs.

The segmentation of chromosome is the most important step in automated analysis of chromosomes [7]. An automated system generally includes the following four steps: (1) image enhancement, (2) segmentation and alignment of the chromosome, (3) computation and selection of characteristics and (4)

chromosomal classification. In these phases, the chromosome segmentation is the most important, because their results can affect the performance of the entire system [8]. Thus, the present work seeks to study and compare approaches for segmentation of fish chromosomes by means of filtering and segmentation based on morphological operations. The filters of the median, gamma and Non Local Means are used to reduce present noise chromosomes and improve image quality.

For chromosomal segmentation the method is based on: Threshold Adaptive and morphological operations, being they, filling of holes, operation of opening and removal of smaller objects. We used the classified SVM (Vector Support Machine) and the k-nearest neighbor algorithm to sort objects between chromosomes and objects with no interest in background. The methods were implemented and tested in a set of images mounted with ground-truth. Finally, an analysis and quantitative comparison of the methods implemented for the segmentation of chromosomes in microscopy images.

This article is organized as follows: This section introduces the introduction on the subject. The II section shows some related work on chromosome segmentation. The III section describes the methods that have been developed and tested, as well as the data set and the validation methods. Section IV presents the results and Section V the conclusions and future work

II. RELATED WORKS

Aln W. and Jane Y. [9] developed an algorithm based on in an adaptive local kernel (KAFCM), a classifier of Probabilistic defuzzification to improve segmentation and classification of chromosomes. This is achieved on a window for each pixel, and compensate for the intensity of the homogeneities caused during the process of generation of images and by the preparation of the physical chromosome itself. O algorithm was tested on a data set available publicly and was compared with the traditional fuzzy clustering algorithm and with the reported results of another adaptive algorithm. Based on these experiments, the proposed system showed comparative results. In addition, the classification results for the proposed method are for defuzzification of standard FCM were compared,

and the the proposed classification method demonstrated an improved overall.

Monika S. et al. [10] presents a method to segment and classify chromosomes for healthy patients using a combination of crowdsourcing, pre processing and learning, said in depth, that the non-specialist multi CrowdFlower is used to separate the chromosomes from the image, which are then straightened and fed in a deep (hierarchical) neural network for classification.

The experiments are performed on 400 real and healthy patients images and the results obtained from a hospital. The results are encouraging and promise to significantly reduce the cognitive aspects of segmentation and karyotyping chromosomes. For some sets of chromosomes, about 95% classification rate for most chromosomes in the set. However, these results only occur for set with better quality images and format. Madian et. al. [11] studied the segmentation of chromosome segmentation. The segmentation was performed taking into account information on the boundary, Otsu threshold, morphological operations and filling holes after binarization. The method uses the detection of edges of objects and the curvature function for detecting cut-off points. The curvature function is obtained by smoothing edges by the median filter. Points detected are concave and convex. The concave points at the edges of the object are used to detect overlap zones, convexes are optional. In this work it was possible to detect more than one overlap in the chromosomes. The algorithm is robust and has been tested on over 350 images with several degrees of overlap. The method in this work obtained a overall accuracy of 96%.

Lijiya A. et al. [12] made a traditional analysis of chromosomes were obtained using grayscale images. At work a majority vote was taken to cromosomica between the techniques and classifiers using diffuse logic for classification of human chromosome images M-FISH. Some techniques of noise removal also are applied to improve the accuracy of ranking segmentation. In some cases the work was precise average of 99% in the segmentation of the chromosomes.

Karvelis et. al. *et. al.* present a method for the segmentation of groups of chromosomes that touch each other and chromosomes superimposed on M-FISH images. Initially, the Transform Watershed is applied at the image and decomposed in regions. Next, gradient paths are calculated from points of concavity high for each region produced. These paths divide the groups of chromosomes from high-concavity points and follow large pixels magnitude of multichannel gradient. Finally, the chromosomal areas are produced to generate the final results. The computation of the gradient path based on multichannel gradient values instead of pale path, which is based on low intensity values, is more robust. The gradient paths deals with overlapping cases of chromosomes, while the pale roads have been able to deal only with groups of chromosomes in contact with each other. To validate the method, we used a reference database of 183 M-FISH images. The algorithm resulted in a success rate of 90.6% for the chromosomes that are touched and of 80.4% for the groups of chromosomes that are superimposed.

Saiyod and Wayalum [13] developed an approach to compute the skeleton of the chromosomes, with the skeleton of the object and it is possible to search the point of intersection. The point of intersection is used to search the candidate points of court. The candidate points are points of curvature found after detecting the edges of the chromosomal images. The cut-off points are found by calculating the Euclidean distance from the point of intersection to the points of curvature. The nearest four points are the points of interest

III. MATERIAL AND METHOD

A. Dataset

The data set used to perform the experiments were assembled from a repository supplied by Laboratory of Evolutionary and Ecological Genetics (LaGEEvo) of UFV's Rio Paranaíba campus. 100 images were selected C in metaphase state. The images were converted for grayscale. For each image there is an image of labels in which each of the chromosomes were segmented manually by the Gimp1 software. Each image have a 1250 x 1250 pixel size and have been saved in a array numpy as an hdf5 file forming a new dataset (Figura 1).

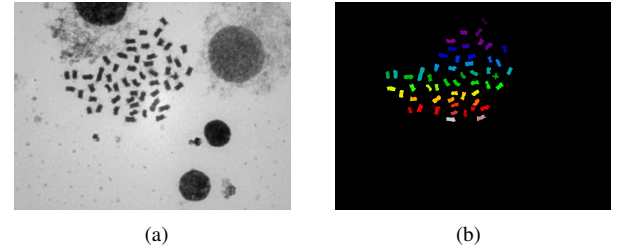


Fig. 1: Images of the data set. (a) Scale image of gray to process (b) Label images

B. Filtering

Chromosome images are generally only of low quality, especially fish, presents noises, low contrast and artifacts, so they must be submitted to a step of preprocessing [13]. The preprocessing step is one of the most important steps in the segmentation process of images, especially in fish chromosomes To test the segmentation between chromosomes and non chromosome was applied four filters: median filter with masks 3 x 3, as it removes ruffles, preserves contours and small details of the chromosomes. Media filter with masks 5 x 5 for smoothing the image [14]. Filter gamma and Non local Means (NLM) with standard deviation of sigma 0.08 and constant (h) 0.6. For an image with noise $v = v(i) | i \in I$, defines defines the operation of the NLM algorithm on a pixel i .

$$NL(v)(i) = \sum_{j \in I} w(i, j) v(j) \quad (1)$$

⁰Available in <https://www.gimp.org>

Where $NL(v)(i)$ and the value to be found of the pixel i ; $w(i, j)$ and the weight of the pixel j in relation to i ; and $v(j)$ the value of j . The weighted average of all pixels in the image with respect to the target pixel to obtain the final value of its filtering [15] [16].

C. Image segmentation in the background and chromosomes

To segment the image, separating the pixels of the chromosomes from the background pixels was used a local threshold or threshold adaptable with block size 47, the block size was chosen empirically, after experiments it was noticed that Very large block sizes the chromosomes tend to to connect. Algorithms of mathematical morphology were applied to improve the quality of the binary image, among they fill holes inside the objects by morphological reconstruction, morphological opening operation, using structuring element in 3 x 3 form, because it can break the isthmus, and tends to smooth out contours. Followed by removal of small objects resulting from the adaptive threshold and connected artifacts at the edge [14]. Figure 2 illustrates the result of the algorithms on the filtered images.

1) *Classification*: Four characteristics were chosen to classify the chromosome particles: the area of the particle (normalized by image size), solidity, maximum intensity and orientation. We divided the set of data in 80% of training images and 20% of images for testing, then the predictions of the classifier SVM (Support Vector Machine) in comparison with the classified parts. The data set was separated in relation to the images, not the objects, since the chromosomes of Each test image should be retrieved to evaluate the result of the segmentation shown in III-B and III-C .

For the classification of chromosome and background the classifier was trained from the 100 metaphase images corresponding to chromosomes or artifacts: nuclei or dust (image segmentation artifact).

Evaluation metrics is commonly used for evaluation those classified in the semantic segments: precision described by Equation 2, recall described by Equation 3, measure (F1-score) as shown in Equation 4 [17] [18]. These metrics are used to evaluate the performance of classifiers in the proposed methods.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1 - Score = \frac{2 \times (Precision \times Recall)}{Precision + Recall} \quad (4)$$

Where $\beta = 1$, TP, FP and FN are respectively *True Positive*, *False Positive* e *False Negative*.

D. Validation

To verify efficiency and compare the methods described in this work and comparing them was used the distance of Hausdorff and the normalized sum of distances (NSD). The

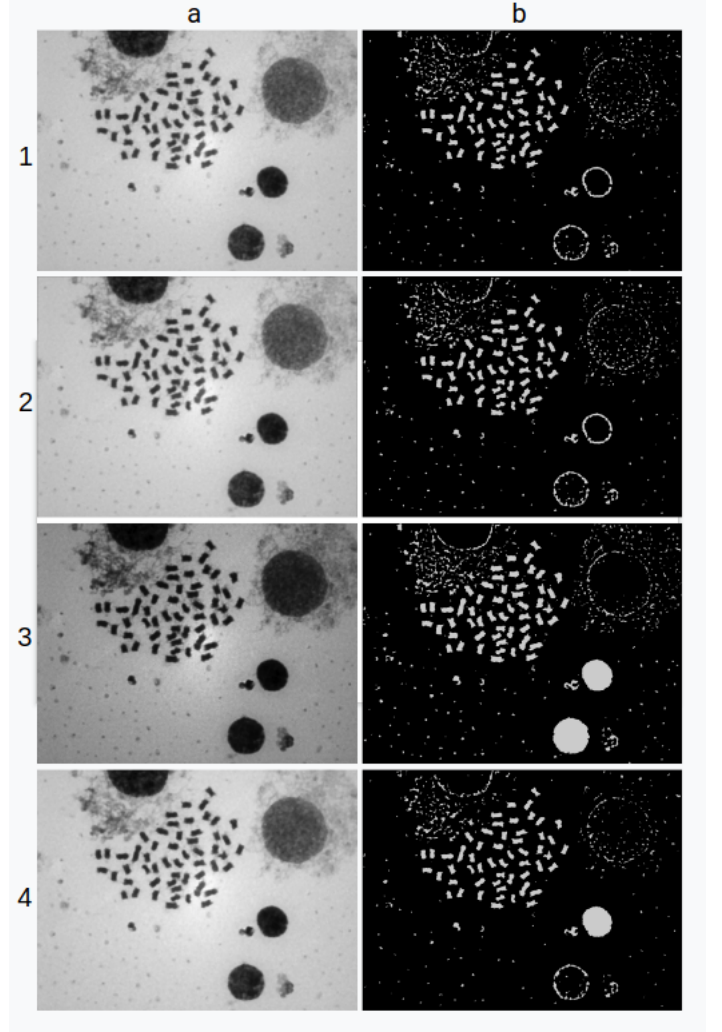


Fig. 2: Segmentation of the image in the background and chromosomes. (1a) Mean filter. (2a) Medium filter. (3a) Gamma filter e (4a) Non-local media filter. The second column (b) shows the segmentation step by means of algorithms of morphology mathematics according to each filter applied.

distance of Hausdorff represents the largest minimum distance between edges of the objects in the segmented image I and the reference R according to Equation 5:

$$Hausdorff(I, R) = \max D(i) : S_i \neq R_i \quad (5)$$

where $D(i)$ is the distance between the pixel i of the object to the edge of the reference object. The value 0 indicates a perfect segmentation, however, the index does not have upper limit [19].

Normalized sum of distances (NSD) defined in Equation 6.

$$NSD(I, R) = \frac{\sum_i I_i \neq R_i * D(i)}{\sum_i D(i)} \quad (6)$$

where $D(i)$ and the distance between the pixel i and the edge of the object of reference. The value 0 indicates a segmentation perfect while 1 indicates that there is no overlap between the segmented cell and to the reference cell.

To evaluate the filtration the noise intensity was calculated the PSNR described by Equation 7.

$$PSNR = 20 \log_{10} \left(\frac{MAX}{\sqrt{MSE}} \right) \quad (7)$$

Where MAX is the maximum possible value of a pixel in a image and MSE (Mean Squared Error) and a measure of quality defined by Equation 8

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \|I(i, j) - K(i, j)\|^2 \quad (8)$$

Where m and n represent the number of rows and columns of pixels of the images, i and j indexes, f the original image and g the filtered image.

IV. RESULTS

All images (100) of the set of images described in Section III-A for the implementation of the All images are filtered according to the procedure described in Section III-B. Then they were segmented in chromosome pixels, segmentation and background artifacts as described in Section III-C. Combinations of filters with the segmentation method. The filtered images and were subjected to the method described in Section III-C1. Combinations of filters with the segmentation and the result of combinations submitted to classifiers SVM and KNN. Table I shows the performance comparison of the classifiers in separating which is chromosomes and which are not to chromosomes for the media filter, Table II comparing for the median filter, Table III for the gamma filter and Table IV for the NLM. As can be seen for the filter all filters the KNN had better accuracy, recall, and better f1-score. Table V, Table VI, Table VII and VIII shows the distance measure of Hausdorff, index of standardized sum of distances and the average intensity of the noise between the result of the described methods compared with the reference image (ground-truth) for the methods. As shown in Table VII the mean for the filter with lower noise intensity in the filtering process was the gamma. From the information in Table VII, to consider that the gamma filter had a better result considering that the other filters consider the measurement of the Hausdorff distance and noise intensity.

Figure ref res shows examples of segmentation results using gamma filter, and KNN. The first column shows the original image in grayscale, the second column shows the result of the method and the third column shows the image considered as ground-truth.

TABLE I: Comparison of the classification results between the SVM and the KNN for the media filter: The first column shows the accuracy, the second recall column, and the third f1-score.

	precision	recall	f1-score
SVM	0.81	0.73	0.77
KNN	0.99	0.99	0.99

TABLE II: Comparison of the classification results between the SVM and KNN for the median filter: The first column shows the second recall column and the third f1-score.

	precision	recall	f1-score
SVM	0.94	0.93	0.93
KNN	0.99	0.99	0.99

TABLE III: Comparison of the classification results between the SVM and KNN for the gamma filter: The first column shows the second recall column and the third f1-score.

	precision	recall	f1-score
SVM	0.88	0.91	0.89
KNN	0.97	0.99	0.98

TABLE IV: Comparison of the classification results between the SVM and KNN for the NLM filter: The first column shows the second recall column and the third f1-score.

	precision	recall	f1-score
SVM	0.87	0.90	0.88
KNN	0.98	0.98	0.98

TABLE V: Mean of the Hausdorff distances, Sum rates normalized distances and the average intensity of the noise computed on the result of both methods described for the media filter and the tested SVM classifiers and KNN

	Hausdorff	NSD	PSNR
SVM	31.55	0.32	34.81
KNN	13.07	0.26	34.81

TABLE VI: Mean of the Hausdorff distances, Sum rates normalized distances and the average intensity of the noise computed on the result of both methods described for the median filter and the SVM and KNN.

	Hausdorff	NSD	PSNR
SVM	52.19	0.48	47.69
KNN	5.69	0.19	47.69

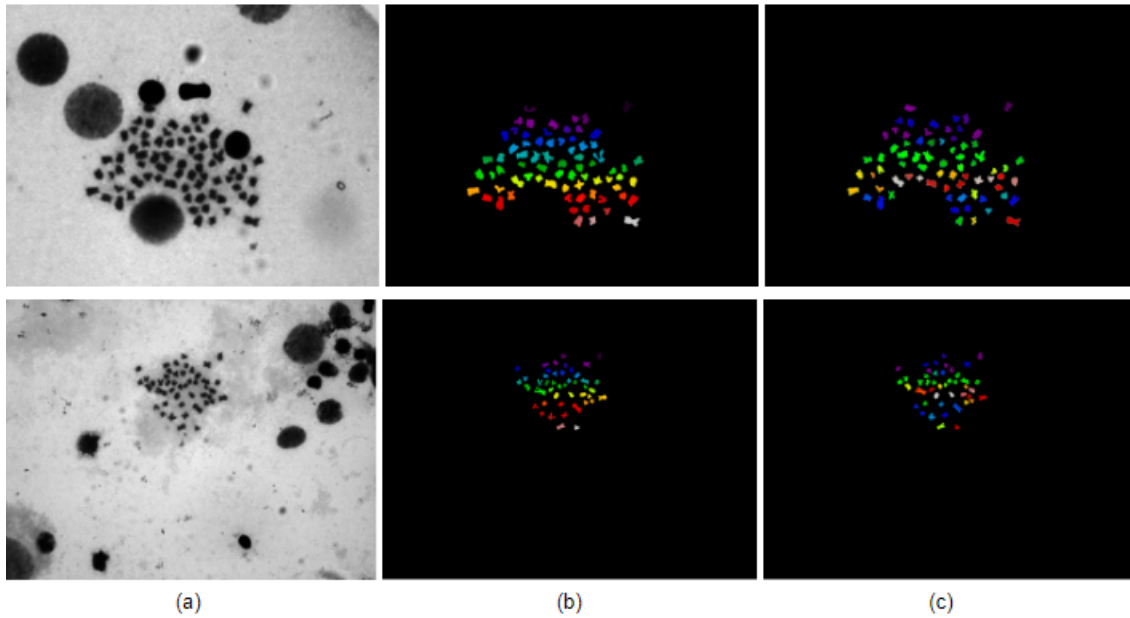


Fig. 3: Segmentation result using the gamma filtering method described in Section ref filters. The first column shows the original image, the second column the resulting image, the third column the reference image. (a) Original image in grayscale; (b) Image resulting from the method by the gamma filter; (c) reference image.

TABLE VII: Mean of the Hausdorff distances, standardized sum of distances and the average intensity of the noise computed on the result of both methods described for the gamma filter and the classifiers tested SVM and KNN.

	Hausdorff	NSD	PSNR
SVM	31.37	0.33	16.87
KNN	5.73	0.17	16.87

TABLE VIII: Mean of the Hausdorff distances, standardized sum of distances and the average intensity of the noise computed on the result of both methods described for the NLM filter and the SVM tested classifiers and KNN

	Hausdorff	NSD	PSNR
SVM	31.86	0.34	36.15
KNN	5.85	0.20	36.15

V. CONCLUSIONS

This article presented a comparison of approaches to segmentation of chromosomes with the media filters, median, median gamma and Non Local Means in microscopy images.

Experiments were carried out using a set of newly constructed data for segmenting fish chromosome. The images were obtained from the Laboratory of Ecological and Evolutionary Genetics of Federal University of Viçosa Rio Paranaíba campus and each image had its chromosomes segmented manually and saved in a h5py dataset. This set of data allows that the result of the experiments to be validated in a pragmatic way.

The algorithms are tested and the results are compared and discussed. It can be seen that the method of the gamma filter combined with the described segmentation of the Section III-C and the KNN classifier showed to be higher than other methods described, 5.73 for distances measurement of Hausdorff and 16.87 at the mean of the noise intensity in images. Like the images of fish chromosomes in metaphase state have a very large amount of noise ratio and does not have much quality, the gamma filter is a solution to this problem.

As future work one can consider testing the using the individually transformed watershed algorithm on chromosomes that touch each other. Another approach using neural networks and deep learning to test the outcome of the segmentation. And finally, to apply a method of classification to for the chromosomes between metacentric, submetacentric central subtotal and acrocentric input for mounting the type of fish. And finally, other different metrics to evaluate the quality of segmentation.

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